

Dynamic Guideline: WHO-Based Tungiasis - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Exactly five WHO/PAHO-WHO official sources synthesized for this tungiasis report

#	WHO/PAHO-WHO source	Date	Relevance to tungiasis
1	Guidelines for the treatment of tungiasis	21 May 2025	Latest disease-specific WHO/PAHO treatment guideline; seven evidence-based recommendations for tungiasis management, including low-viscosity dimethicone and avoidance of harmful practices PAHO/WHO, Guideline Central .
2	WHO Tungiasis fact sheet	28 April 2023	WHO clinical and public-health summary covering cause, morbidity, diagnosis/treatment context, NYDA® dimeticone, sterile extraction, tetanus prevention, and One Health control WHO fact sheet .
3	WHO skin neglected tropical diseases framework	2022	WHO framework for integrated skin-NTD management; WHO notes that tungiasis is included to promote integration across skin diseases WHO fact sheet .
4	WHO neglected tropical diseases	2021–2030; progress	WHO road-map context for surveillance, visibility, and programmatic control; WHO

#	WHO/PAHO-WHO source	Date	Relevance to tungiasis
	road map 2021–2030 / progress reporting	document reviewed to 2026	reported tungiasis remains underreported and that a tungiasis guideline was released among recent NTD publications WHO road map progress document .
5	World Health Assembly resolution on skin diseases as a global public-health priority	2025	Supports sustainable prevention strategies for skin diseases, including WASH and One Health approaches relevant to tungiasis transmission reduction WHA78 resolution .

Important scope note: The WHO publication “**Meaningful youth engagement in a WHO guideline development process: case study of WHO Abortion care guideline (2022)**” is a WHO guideline-development methodology case study, not a tungiasis guideline. It may inform future inclusive tungiasis guideline updates, especially because tungiasis heavily affects children, but the available sources show no direct link between this youth-engagement brief and the tungiasis guideline [WHO webinar](#), [WHO publication PDF](#).

Latest disease-specific guideline

- **Guideline title:** *Guidelines for the treatment of tungiasis*
- **Publisher:** Pan American Health Organization / World Health Organization
- **Publication date:** 21 May 2025 [Guideline Central](#)
- **Bibliographic citation:** Pan American Health Organization. *Guidelines for the treatment of tungiasis*. Washington, D.C.: PAHO; 2025 [PAHO IRIS](#).
- **ISBN:** 978-92-75-13001-8 [PAHO IRIS](#).
- **Target population:** People with tungiasis in endemic, marginalized, resource-poor communities, especially in South America, the Caribbean, and sub-Saharan Africa; children are a particularly affected group [PAHO/WHO](#), [WHO fact sheet](#).
- **Scope and objectives:** Evidence-based treatment and management of tungiasis caused by the sand flea *Tunga penetrans*, with recommendations for effective therapy, avoidance of harmful practices, and community-based approaches to reduce morbidity and transmission [PAHO/WHO](#).

1.2 Key Recommendations

GRADE note: Guideline Central confirms that the WHO/PAHO tungiasis recommendations are GRADE-based and use the GRADE certainty categories High, Moderate, Low, and Very Low

Guideline Central. The supplied sources do not expose the exact official certainty and strength for each individual recommendation. Therefore, the table below provides a **report-derived GRADE assessment** based on the available guideline summaries and supporting evidence, not a verbatim extraction of the full WHO evidence-to-decision tables.

Recommendation	GRADE Evidence Quality	Strength
Diagnose tungiasis clinically in endemic or exposure-compatible settings by identifying characteristic sand-flea lesions; no routine laboratory test is required for typical cases.	Low	Strong
Use topical low-viscosity dimeticone oils, especially NYDA®, as the preferred treatment where available. WHO describes NYDA® as the only product shown to have high efficacy and safety in the available WHO fact-sheet summary WHO fact sheet .	Moderate	Strong
For severe tungiasis, avoid non-professional mechanical extraction because of risk of secondary infection and physical trauma; if extraction is used, it should be sterile and performed by trained personnel, followed by wound care and tetanus-status review WHO fact sheet , PAHO/WHO .	Low	Strong
Discourage hydrogen peroxide as routine tungiasis treatment because guideline summaries identify it among practices discouraged by WHO/PAHO PAHO/WHO .	Very Low	Conditional
Discourage potassium permanganate as routine treatment, particularly when safer evidence-supported options are available PAHO/WHO .	Very Low	Conditional
Consider coconut oil combined with neem oil as an alternative in resource-limited settings when preferred treatment is unavailable PAHO/WHO .	Low	Conditional
Consider topical ivermectin only when preferred treatments are unavailable; older randomized evidence suggests ivermectin and other antihelminthics have little or no efficacy against embedded sand fleas, so it should not displace dimeticone where dimeticone	Very Low	Conditional

Recommendation	GRADE Evidence Quality	Strength
is accessible Guideline Central , PMC dimeticone comparison study .		
Implement community education and community-based access to proper treatment to reduce harmful extraction practices, stigma, and delayed care Guideline Central .	Low	Strong
Use a One Health control approach: treat affected people, improve household environments, spray or seal floors where feasible, promote daily foot washing with soap, and treat infected household animals WHO fact sheet .	Low	Strong

1.3 Guideline Development Methodology

- **Evidence review process:** The WHO/PAHO guideline is described as a comprehensive technical guide grounded in the latest scientific research and containing seven evidence-based recommendations [PAHO/WHO](#). Guideline Central identifies the recommendations as GRADE-based [Guideline Central](#).
- **Consensus method:** The supplied summaries confirm a PAHO Guidelines Development Group was involved, but the detailed voting or consensus rules were not available in the supplied source extracts [SciDev.Net](#).
- **Conflict of interest management:** The supplied summaries did not expose the full conflict-of-interest section.
- **Stakeholder engagement:** No supplied tungiasis source documents youth engagement in the tungiasis guideline-development process. WHO's youth-engagement evidence brief is directly about the 2022 WHO Abortion care guideline and should be cited only as general WHO methodology, not as tungiasis-specific evidence [WHO publication PDF](#).

II. Evidence Summary After WHO Latest Guideline Release (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy

Evidence was synthesized from the supplied summaries using searches focused on:

- WHO/PAHO tungiasis guideline, diagnosis, treatment, and control documents.
- Tungiasis randomized trials and systematic reviews from 2024–2026.
- Dimeticone/NYDA®, sodium carbonate, ivermectin, neem/coconut oil, potassium permanganate, phenothrin, and mechanical extraction.
- Authoritative diagnostic references such as CDC DPDx.
- WHO guideline-development methodology documents, including youth-engagement material, assessed for relevance.

Databases and sources searched or represented in the supplied summaries

- WHO official publications and fact sheets.
- PAHO/WHO document repository and PAHO IRIS.
- Guideline Central summary of WHO/PAHO guideline.
- PubMed / PMC.
- PLOS Neglected Tropical Diseases.
- BMJ Open.
- Lancet Infectious Diseases systematic-review evidence as cited in BMJ Open.
- CDC DPDx.
- WHO and UN youth-engagement methodology pages.

Inclusion criteria

- Human tungiasis diagnosis, treatment, prevention, control, or management.
- WHO/PAHO or other authoritative institutional guidance.
- Randomized trials, non-inferiority trials, systematic reviews, meta-analyses, and major observational burden/risk-factor studies.
- Evidence available up to **29 April 2026**.

Exclusion criteria

- Non-tungiasis NTD guidance unless directly relevant to skin-NTD integration or One Health management.
- Youth-engagement publications as clinical evidence for tungiasis, because no direct tungiasis link was identified.
- Sources without relevance to diagnosis, treatment, prevention, control, or guideline methods.

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
Suzuki et al., 2026: dimeticone oils versus sodium carbonate solution in Kenya Suzuki et al., 2026	Randomized non-inferiority trial	Compared NYDA® dimeticone oils with 5% sodium carbonate solution. Protocol specified school children with tungiasis, one treatment per foot, monitoring of flea viability, inflammation, symptoms, and adverse events for seven days Protocol . Final effect estimates were not visible in the supplied summaries.	Moderate
Suzuki et al., 2024: pilot randomized cohort trial, Homa Bay County, Kenya PLOS NTD , PubMed	Randomized, observer-blinded pilot cohort trial	23 children aged 3–13 years; 126 flea infections. Day-7 dead fleas: 87% with NYDA® versus 64% with 5% sodium carbonate, P = 0.01. No significant difference in inflammation score or pain/itchiness in supplied summary. Small pilot; predates the WHO guideline but supports dimeticone as preferred treatment.	Low
Randomized parallel-group study: 3% phenothrin-containing jigger lotion versus potassium	Randomized parallel-group trial	Relevant because WHO/PAHO discourages potassium permanganate, while this study directly compares it with phenothrin lotion. Detailed effect estimates	Low

Study	Design	Key Findings	GRADE Quality
permanganate, Kenya Phenothrin trial		were not available in supplied summaries.	
Girma et al., 2025: prevalence and risk factors for tungiasis in Africa	Systematic review and meta-analysis	Identified in a recent trial reference list as a 2025 BMC Infectious Diseases systematic review/meta-analysis. Detailed pooled estimates were not available in the supplied summaries Phenothrin trial .	Low
Abrha et al., 2021: clinical interventions for tungiasis	Systematic review	Identified eight randomized controlled trials; found the greatest promise for dimeticones for treatment and a coconut oil-based lotion for prevention, though the prevention lotion was no longer commercially available BMJ Open PDF . Predates guideline but remains central background evidence.	Moderate
CDC DPDx Tungiasis [CDC D			